

Integrating a block expansion over noncritical parameters

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Abstract

An abstract wrapper for the "equal-exponent block times noncritical coordinates" situation (Astra #4, rank 1). If a parametrised family $Z_w(n)$ satisfies $|Z_w(n) - n^{-p}(A(w) \log n + B(w))| \leq K(w)n^{-q}(1 + \log n)$ for $n \geq 1$ ($q = p + \delta$), $|Z_w(n)| \leq G(w)$ for $n < 1$, $0 < t(w) \leq T$, and the envelope $t^{-q}(G + K + |A| + |B|)$ is μ -integrable (the strict gap), then

$$\int Z_w(Nt(w)) d\mu = N^{-p}(\bar{A} \log N + \bar{B}) + O(N^{-q}(1 + \log N)),$$
$$\bar{A} = \int t^{-p} A d\mu, \quad \bar{B} = \int t^{-p} (B + A \log t) d\mu.$$

The region $Nt(w) < 1$ is handled by subtracting the full local expression and the elementary bound $n^{-p}|\log n| \leq n^{-q}/\delta$. The wrapper knows nothing about kernels, decompositions or derivatives. Zero `sorry/axiom`.

1 The things formalised

```
theorem rpow_mul_abs_log_le (x : ℝ) (hx : 0 < x) (hx1 : x < 1) (h : 0 < ) :
  x ^ * |Real.log x| ^ 1 /
theorem rpow_neg_le_shift (t T p : ℝ) (ht : 0 < t) (htT : t < T) (h : 0 < ) :
  t ^ (-p) T ^ * t ^ (-p + )
noncomputable def logShiftConst (T : ℝ) : ℝ := 1 / + T ^ * max 0 (Real.log T)
theorem rpow_neg_mul_abs_log_le_shift ... :
  t ^ (-p) * |Real.log t| logShiftConst T * t ^ (-p + )
noncomputable def paramConst (T : ℝ) : ℝ := 1 + max 0 (Real.log T) + 1 /
theorem param_pointwise_bound (Z : ℝ → ℝ) (A B K G t T p N : ℝ) (h : 0 < ) (hp : 0 < p)
  (ht : 0 < t) (htT : t < T) (hN : 1 < N) (hK : 0 < K) (hG : 0 < G)
  (hexp : ℝ, 1 < n →
    |Z n - n ^ (-p) * (A * Real.log n + B)| ≤ K * n ^ (-p + ) * (1 + Real.log n))
  (hsmall : ℝ, 0 < n → n < 1 → |Z n| ≤ G) :
  |Z (N * t) - N ^ (-p) * (t ^ (-p) * A * Real.log N + t ^ (-
p) * (B + A * Real.log t))|
  N ^ (-p + ) * (1 + Real.log N)
  * (paramConst T * (t ^ (-p + ) * (G + K + |A| + |B|)))
theorem param_integration {Ω : Type*} [MeasurableSpace Ω] ( : Measure Ω)
  (Z : Ω → ℝ) (A B K G t : Ω → ℝ) (T p : ℝ) (h : 0 < ) (hp : 0 < p) (hT : 0 < T)
  (htm : Measurable t) (hAm : AESTronglyMeasurable A) (hBm : AESTronglyMeasurable B)
  (hZm : N : ℝ, AESTronglyMeasurable (fun w => Z w (N * t w)))
  (ht : ℝ, 0 < t < T)
  (hKG : ℝ, 0 < K < G)
  (hexp : ℝ, 1 < n →
```

```

|Z w n - n ^ (-p) * (A w * Real.log n + B w)| K w * n ^ (-
(p + )) * (1 + Real.log n))
(hsmall : w , n, 0 < n → n < 1 → |Z w n| G w)
(hE : Integrable (fun w => t w ^ (-p + )) * (G w + K w + |A w| + |B w|)) ) :
C : , N : , 1 N →
| ( w, Z w (N * t w) )
- N ^ (-p) * (( w, t w ^ (-p) * A w ) * Real.log N
+ w, t w ^ (-p) * (B w + A w * Real.log (t w)) ) |
C * N ^ (-p + )) * (1 + Real.log N)

```

2 Mathematics

With $n = Nt(w)$ the full local expression $N^{-p}(t^{-p}A \log N + t^{-p}(B + A \log t))$ is exactly $n^{-p}(A \log n + B)$, so the pointwise deviation is the block remainder at n . On $n \geq 1$ it is bounded by $Kn^{-q}(1 + \log n)$ and $1 + \log n = 1 + \log N + \log t \leq (1 + \log^+ T)(1 + \log N)$. On $n < 1$ we bound $|Z_w(n)| \leq G$, $n^{-p} \leq n^{-q}$ and $n^{-p}|\log n| = n^{-q}n^\delta|\log n| \leq n^{-q}/\delta$ (from Mathlib's $|x \log x| < 1$ on $(0, 1]$ applied to $x = n^\delta$), and use $1 \leq n^{-q}$ to absorb G . Both regions give $n^{-q} \cdot \text{const} \cdot (G + K + |A| + |B|) = N^{-q}t^{-q}(\dots)$, and integrating against μ costs exactly the envelope moment.

The coefficient integrals $\bar{A}, \bar{B}$ are finite because $t^{-p} \leq T^\delta t^{-q}$ and $t^{-p}|\log t| \leq (1/\delta + T^\delta \log^+ T) t^{-q}$ on $(0, T]$, so both integrands are dominated by the same envelope. The strict gap $p_\ell > p + \delta$ is what makes $\int t^{-q} d\mu$ finite in the application ($\mu = w^h dw$ on the noncritical box): this wrapper takes the envelope moment as its single integrability hypothesis.

3 Paper-facing meaning

This is the general mechanism by which the $d = 2$ equal-exponent block theorem (units 86–89) extends to the mixed situation of `thm:TaylorTree`: a block of two coordinates attaining the minimal exponent, plus noncritical coordinates with strictly larger exponents. The next unit instantiates Z_w by the parametrised chart block, supplying measurable $A(w), B(w), K(w), G(w)$ from the anchored decomposition with uniform constants, and the envelope moment from the strict gap.

4 Lean notes

- `Real.abs_log_mul_self_lt` gives $|x \log x| < 1$ on $(0, 1]$ with constant 1; the sharp $1/e$ is not needed, so $n^{-p}|\log n| \leq n^{-q}/\delta$ suffices.
- `Real.rpow_le_rpow_of_exponent_ge` (base in $(0, 1]$, exponents reversed) gives both $n^{-p} \leq n^{-q}$ and $1 = n^0 \leq n^{-q}$.
- `mul_le_mul_of_nonneg_right` wants $0 \leq c$, not $1 \leq c$; a `by linarith` in place of a misnamed hypothesis is cheaper than a rename.
- `positivity` cannot see $0 \leq K w$ from a conjunction hypothesis; extract the components and use `mul_nonneg` explicitly.
- The final assembly is `integral_sub + integral_add + integral_const_mul + norm_integral_le_of_norm_le`; integrability of the scaled block itself is obtained from the pointwise bound as $(f - g) + g$.